Mahalingam Ramasamy

Software Engineer

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Summary

Software Engineer with 2+ years building scalable backend systems in Node.js (NestJS) and Java (Spring Boot) on AWS. Delivered real-time WebSocket services, event-driven pipelines, payment integrations and a multi-provider LLM layer with automatic failover. Strong problem solver with a solid grounding in data structures and algorithms.

Experience

Ekko Medical Solutions Pvt Ltd **Oct 2025 – Present**

Software Developer (Backend) *Chennai, India*

- Engineered a horizontally scalable **real-time transcription service** for an AI medical scribe (**WebSockets, Redis Streams, PostgreSQL**), delivering **sub-second** transcripts for **5-hour sessions** with auto-recovery.
- Cut clinical-note generation **latency by 50%** (50s to 25s) by redesigning the flow as an **event-driven, asynchronous pipeline** with a lightweight refinement LLM.
- Designed a **provider-agnostic LLM service layer** across 3 providers (OpenAI, Gemini, Anthropic) serving **15+ tasks**, with per-task routing and **same-request failover** so a provider outage doesn't fail the request.
- Built **Cashfree** subscription billing with **UPI/eNACH autopay across 7 plans**, HMAC-verified webhooks, idempotent processing and hourly reconciliation.

ROX Hi-TECH **Jun 2024 – Oct 2025**

Integration Engineer (Full Stack) *Chennai, India*

- Developed a full-stack HR platform (**Java, Spring Boot, React**) for **TNEB**, Tamil Nadu's state electricity board, digitizing leave, transfer and performance-review requests for **100,000+ employees**.
- Replaced paper-based approvals with a configurable approval engine across **6+ request types**, cutting turnaround by **70%** and letting new workflows launch through configuration instead of code changes.
- Built **Spring Boot REST APIs** integrating **SAP ERP** via **OData** to sync employee master data, eliminating duplicate data entry between the HR platform and SAP.
- Implemented **role-based access control** across **3 user roles** with **Spring Security**, enforcing approvals at the API layer so each role can view and act only on requests within its scope.

Projects

Shoppy – E-Commerce Platform | *Java, Spring Boot, Next.js, PostgreSQL, Stripe, AWS* **GitHub**

- Integrated **Stripe** payments with signature-verified, **idempotent webhooks**, guaranteeing exactly-once order creation even when Stripe retries events.
- Secured APIs with **Spring Security, JWT** and role-based access, and pushed live product updates to the Next.js storefront over **WebSockets (STOMP)**.
- Wrote **54 automated tests** (JUnit 5, Mockito, Testcontainers) reaching **80% coverage**, run in CI on every push.
- Automated deployments to **AWS** with CodePipeline and **Elastic Beanstalk**, serving data from RDS PostgreSQL and assets from S3 behind CloudFront.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, SQL

Backend: Node.js, NestJS, Spring Boot, FastAPI, REST APIs, WebSockets, SSE, Celery

Databases: PostgreSQL, pgvector, Redis, Prisma

Cloud: AWS (ECS Fargate, ALB, Lambda, S3, CloudFront, SES, VPC, Secrets Manager, CloudWatch), GCP

DevOps: Terraform, Docker, GitHub Actions, CI/CD

CS Fundamentals: Data Structures & Algorithms, Problem Solving, System Design

Frontend: React, Next.js

Education

SASTRA Deemed University **Jun 2020 – Jun 2024**

Bachelor of Technology in Computer Science and Business Systems *Thanjavur, India*

- **Relevant Coursework:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks